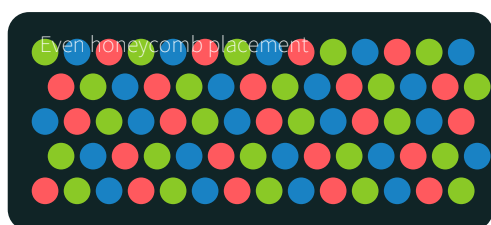


LEDScreenShader

Start Guide / スタートガイド

Version 2.0.0 | Native HLSL shader for realistic LED panels in Unity

This PDF is the offline documentation included with the Asset Store package. It contains an English setup guide, shader property reference, migration notes, troubleshooting, and a Japanese quick guide.



Procedural LED

Honeycomb, stripe, and rectangle patterns can be generated without an LED texture. Honeycomb placement is designed to avoid visible color bias.

Supported targets: HDRP and URP. Built-in Render Pipeline includes a simplified fallback shader only and is outside the official support target.

Table of Contents

Numbered sections for Asset Store review

1. Overview
2. Render Pipeline Support
3. Quick Setup Tutorial
4. Illustrated Workflow
5. Material Property Reference
6. Migration From Legacy Shader Graph Materials
7. Troubleshooting and Recommendations
8. Editor Script Reference
9. Japanese Quick Guide / 日本語クイックガイド
10. Japanese Migration and Troubleshooting / 日本語移行ガイド

1. Overview

LEDScreenShader renders realistic LED panels in Unity using the native shader Ilcheesell/LEDScreen. It displays a texture or RenderTexture through an LED subpixel pattern, then applies emission intensity, distance fading, procedural LED layout, and optional cabinet seams.

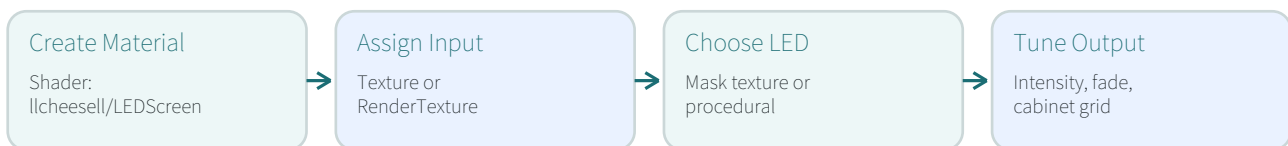
- No runtime scripting is required to use the asset.
- The previous Shader Graph implementation has been replaced by the native HLSL shader in version 2.0.0.
- A migration tool is included for legacy materials and Missing Shader cases caused by older package updates.

2. Render Pipeline Support

Pipeline	Status	Notes
HDRP	Supported	Tested in Unity 2021 and Unity 6000.3.15.
URP	Supported	Tested in Unity 2021 and Unity 6000.3.15.
Built-in	Fallback only	Simplified fallback shader; not officially supported.

3. Quick Setup Tutorial

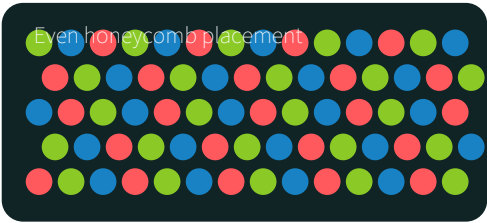
1. Create a new Material in your Unity project.
2. Set the shader to Ilcheesell/LEDScreen.
3. Assign a texture or RenderTexture to Input Screen Texture.
4. Select an LED Mask texture, or enable Procedural LED and choose a pattern.
5. Adjust LED Columns and LED Rows for the desired density.
6. Adjust Intensity for the scene exposure and Bloom settings.
7. Enable Distant Fader if fine LED detail creates moire at camera distance.
8. Enable Cabinet Grid when cabinet seams should be visible.



4. Illustrated Workflow



The input texture provides the image. The LED mask or procedural layout determines the subpixel pattern. The native shader combines them with fade controls, emission, optional surface material, and optional cabinet seams before rendering the final screen.



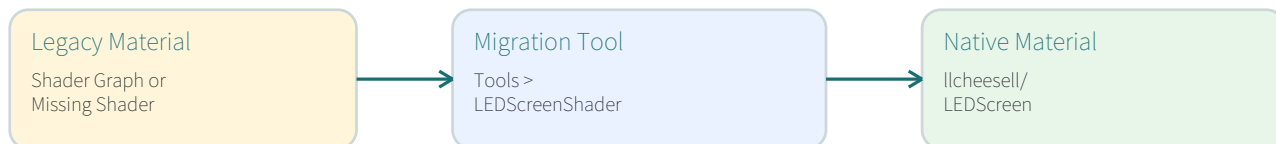
Procedural LED

Honeycomb, stripe, and rectangle patterns can be generated without an LED texture. Honeycomb placement is designed to avoid visible color bias.

5. Material Property Reference

Group	Property	Purpose
Input	Input Screen Texture	Image, video texture, or RenderTexture displayed on the panel.
LED	LED Mask / Procedural LED	Controls RGB subpixel layout or generates LED dots in shader.
Brightness	Emission Color / Intensity	Controls HDR output and Bloom response.
Fade	Fade Start / End / Bias	Reduces moire when LED detail becomes too dense on screen.
Cabinet	Cabinet Grid	Adds panel seams, depth, and subtle brightness variation.
Surface	Base / Normal / Mask	Optional PBR surface material for the screen body.

6. Migration From Legacy Shader Graph Materials



Input Texture / video input is copied to Input Screen Texture (`_InputTex`).

- When legacy LEDScreenShader materials are found, the Unity Editor shows a migration prompt once.
- You can run the tool manually from Tools > LEDScreenShader > Migrate Legacy Materials to Native Shader.
- The tool detects previous HDRP Shader Graph, URP Shader Graph, and legacy Built-in shader references by known shader names and GUIDs.
- Materials that appear as Missing Shader after updating from an older GitHub package can also be detected when the old shader GUID is known.
- The old input/video texture is assigned to Input Screen Texture (`_InputTex`). Compatible LED textures, LED tiling, input texture tiling/offset, emission color, and intensity are copied where possible.

Important: Review migrated materials visually. The tool copies compatible settings, but exposure, Bloom, and render pipeline lighting can change the final look.

7. Troubleshooting and Recommendations

- If the screen is too dark, increase Intensity and check exposure or Bloom settings.
- If moire appears, enable Distant Fader and tune Fade Start / End.
- If the material shows Missing Shader after an update, run the migration tool.
- Use Linear Color Space when possible. Gamma Color Space can clamp bright areas more easily.

8. Editor Script Reference

This asset does not require users to write code. The included scripts are Editor-only helpers used for material UI and migration.

Script	Purpose
LEDShaderGUI	Custom Material Inspector for Ilcheesell/LEDScreen. It groups shader properties into foldable sections.
LEDShaderMigration	Scans and migrates legacy materials to the native shader. It runs only in the Unity Editor.

Package Notes

- The supported shader in version 2.0.0 is Ilcheesell/LEDScreen.
- HDRP and URP are the supported render pipelines.
- The Built-in Render Pipeline fallback is included only as a simplified compatibility path.
- The previous Shader Graph implementation is not included in the main package.

9. 日本語クイックガイド

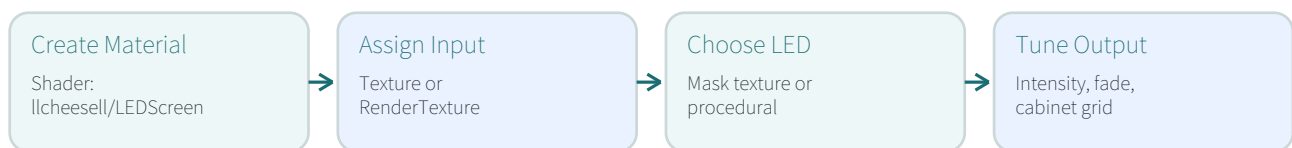
LEDScreenShaderは、Unity上で高品質なLEDパネル表現を行うネイティブHLSLシェーダーです。Version 2.0.0では、従来のShader Graph実装から Ilcheesell/LEDScreen へ移行しました。

対応レンダーパイプライン

- HDRP: 対応、動作確認済み。
- URP: 対応、動作確認済み。
- Built-in Render Pipeline: 簡易Fallbackのみ同梱。公式サポート対象外です。

セットアップ手順

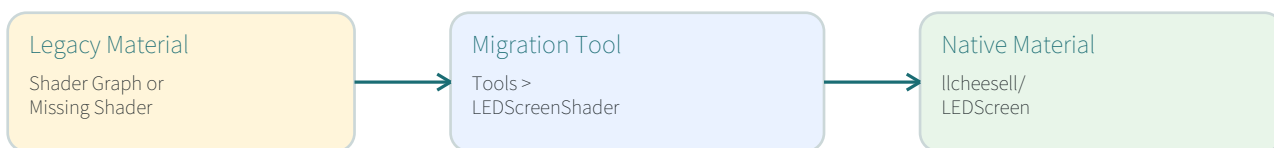
1. 新規Materialを作成します。
2. Shaderを Ilcheesell/LEDScreen に設定します。
3. Input Screen Texture に表示したいTextureまたはRenderTextureを指定します。
4. LED Maskを選ぶか、Procedural LEDを有効にしてパターンを選択します。
5. LED Columns / LED RowsでLED密度を調整します。
6. Intensityをシーンの露出やBloomに合わせて調整します。
7. モアレが出る場合はDistant Faderを有効にします。



10. 日本語移行ガイド

旧Shader Graph版のマテリアルが見つかった場合、Unity

Editor上で移行プロンプトが表示されます。手動で実行する場合は Tools > LEDScreenShader > Migrate Legacy Materials to Native Shader を使用してください。



Input Texture / video input is copied to Input Screen Texture (`_InputTex`).

- 旧Input/Videoテクスチャは、新ShaderのInput Screen Texture (`_InputTex`) に割り当てます。
- LEDテクスチャ、LEDタイリング、Input TextureのTiling/Offset、Emission Color、Intensityは可能な範囲で引き継ぎます。
- GitHubパッケージ更新後にMissing Shaderになったマテリアルも、既知の旧Shader GUIDから検出できます。
- 移行後は見た目を確認し、Intensity、Bloom、Distant Faderをシーンに合わせて調整してください。

主なプロパティ

- Input Screen Texture: パネルに表示する映像・画像・RenderTexture。
- LED Mask / Procedural LED: LEDサブピクセルの配置。
- Intensity: HDR発光とBloomの強さ。
- Distant Fader: 距離に応じてLEDディテールを抑え、モアレを軽減。
- Cabinet Grid: LEDキャビネットの継ぎ目や明暗差を追加。